

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Conditional Probability and Multiplication Theorem on Probability

If A and B are two events such that $P(A) \neq 0$ and $P(B) \neq 1$, then $P(\bar{A}/\bar{B})$ is equal to

- (1) $1 - P(A/B)$
- (2) $1 - P(A/\bar{B})$
- (3) $\frac{1 - P(A \cup B)}{P(\bar{B})}$
- (4) $\frac{P(\bar{A})}{P(\bar{B})}$

Q2 - Conditional Probability and Multiplication Theorem on Probability

A coin is tossed three times in succession. If E is the event that there are atleast two heads and F is the event in which first throw is a head, then $P(E/F)$ is equal to

- (1) $3/4$
- (2) $3/8$
- (3) $1/2$
- (4) $1/8$

Q3 - Conditional Probability and Multiplication Theorem on Probability

A bag contains 3 red, 6 white and 7 black balls. Two balls are drawn one-by-one, then the probability that first ball is white and second ball is blue when first drawn ball is not replaced in the bag, is

- (1) $7/40$
- (2) $9/40$
- (3) $11/40$
- (4) $13/40$

Q4 - Conditional Probability and Multiplication Theorem on Probability

If two events A and B are such that $P(\bar{A}) = 0.3$, $P(B) = 0.4$ and $P(A\bar{B}) = 0.5$ Then, $P[B/(A \cup \bar{B})]$ is

- (1) $1/2$

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(2) $\frac{1}{3}$

(3) $\frac{1}{4}$

(4) $\frac{1}{5}$

Q5 - Conditional Probability and Multiplication Theorem on Probability

A bag contains 2 black, 4 white and 3 red balls. One ball is drawn at random from the bag and kept aside. From the remaining balls another ball is drawn and kept aside the first. This process is repeated till all the balls are drawn. Then, probability that the balls are in sequence of 2 black, 4 white and 3 red is

(1) $\frac{1}{1260}$

(2) $\frac{1}{7560}$

(3) $\frac{1}{210}$

(4) None of these

Q6 - Conditional Probability and Multiplication Theorem on Probability

There are 6 red balls and 6 green balls in a bag. Five balls are drawn out at random and placed in a red box. The remaining seven balls are put in a green box. If the probability that the number of red balls in the green box plus the number of red balls is not a prime number, is $\frac{p}{q}$, where p and q are relatively prime. Then, the value of $(p + q)$ is

(1) 35

(2) 36

(3) 37

(4) 38

Q7 - Independent Events

Whenever horses a , b and c race together, their respective probabilities of winning the race are 0.3, 0.5 and 0.2, respectively. If they race three times the probability that 'the same horse wins all the races' and the probability that a , b and c each wins one race, are respectively (assume no dead heat), is

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(1) $4/25, 9/50$

(2) $16/100, 3/100$

(3) $12/50, 15/50$

(4) $10/50, 8/50$

Q8 - Independent Events

Two cubes have their faces painted either red or blue. The first cube has five red faces and one blue face.

When the two cubes are rolled simultaneously, the probability that the two top faces show the same colour is $(1/2)$. Number of red faces on the second cube, is

(1) 1

(2) 2

(3) 3

(4) 4

Q9 - Independent Events

Four persons independently solve a certain problem correctly with probabilities $1/2, 3/4, 1/4, 1/8$. Then, the probability that the problem is solved correctly by atleast one of them, is

(1) $\frac{235}{256}$

(2) $\frac{21}{256}$

(3) $\frac{3}{256}$

(4) $\frac{253}{256}$

Q10 - Independent Events

A man alternately tosses a coin and throws a die beginning with the coin. The probability that he gets a head in the coin before he gets 5 or 6 in the die, is

(1) $3/4$

(2) $1/2$

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(3) $1/3$

(4) None of these

Q11 - Independent Events

A pair of fair dice is rolled together till a sum of either 5 or 7 is obtained. Then, the probability that 5 comes before 7, is

(1) $1/5$ (2) $2/5$ (3) $4/5$

(4) None of these

Q12 - Independent Events

A biased coin with probability P , $0 < P < 1$, of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $2/5$, then the value of P is

(1) $1/4$ (2) $1/6$ (3) $1/3$ (4) $1/2$

Q13 - Independent Events

Two numbers a and b are selected from the set of natural numbers, then the probability that $a^2 + b^2$ is divisible by 5, is

(1) $9/25$ (2) $7/18$ (3) $11/36$ (4) $17/81$

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Q14 - Law of Total Probability

Three groups A , B and C are contesting for positions on the Board of Directors of a company. The probabilities of their winning are 0.5, 0.3 and 0.2, respectively. If the group A wins, the probability of introducing a new product is 0.7 and the corresponding probabilities for groups B and C are 0.6 and 0.5, respectively. Then, the probability that the new product will be introduced, is

- (1) 0.63
- (2) 0.35
- (3) 0.18
- (4) 0.10

Q15 - Law of Total Probability

There are n bags such that i th bag ($1 \leq i \leq n$) contains i black and 2 white balls. Two balls are drawn from a randomly selected bag out of given n bags. Then, the probability that the both drawn balls are white, is

- (1) $\frac{1}{n}$
- (2) $\frac{1}{n+1}$
- (3) $\frac{1}{n+2}$
- (4) $\frac{n}{n+2}$

Q16 - Law of Total Probability

A is one of the 6 horses entered for a race and is to be ridden by one of two jockeys B or C . It is 2 to 1 that B rides A , in which case all the horses are equally likely to win, if C rides A , his chance is trebled, then the probability A wins the race, is

- (1) $\frac{13}{18}$
- (2) $\frac{5}{18}$
- (3) $\frac{7}{18}$
- (4) None of these

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Q17 - Law of Total Probability

Let A consists of 3 good and 2 defective articles. Let B consists of 4 good and 1 defective article. A new to C is formed by taking 3 articles from A and 2 from B . The probability that an article chosen at random from C is defective, is

- (1) $\frac{1}{3}$
- (2) $\frac{2}{5}$
- (3) $\frac{8}{25}$
- (4) None of these

Q18 - Bayes Theorem

A bag A contains 2 white balls, 3 red balls and bag B contains 4 white balls, 5 red balls. One ball is drawn at random from a randomly chosen bag and is found to be red. The probability that it was drawn from bag B , was

- (1) $\frac{5}{14}$
- (2) $\frac{5}{16}$
- (3) $\frac{5}{18}$
- (4) $\frac{25}{52}$

Q19 - Bayes Theorem

A letter is known to have come either from 'KRISHNAGIRI' or 'DHARMAPURI'. On the post mark only the two consecutive letters 'RI' are visible. Then, the chance that it came from 'KRISHNAGIRI' is

- (1) $\frac{3}{5}$
- (2) $\frac{2}{3}$
- (3) $\frac{9}{14}$
- (4) $\frac{5}{14}$

Q20 - Bayes Theorem

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A purse contains 100 coins of unknown value, a coin drawn at random is found to be a rupee. The chance that it is the only rupee in the purse, is (assume all numbers of rupee coins in the purse to be equally likely)

- (1) $\frac{1}{5050}$
- (2) $\frac{2}{5151}$
- (3) $\frac{1}{4950}$
- (4) $\frac{2}{4950}$

Q21 - Bayes Theorem

On a Saturday night, 20% of all drivers in the USA are under the influence of alcohol. The probability that a driver under the influence of alcohol will have an accident is 0.001. The probability that a sober driver will have an accident is 0.0001. If a car on a Saturday night smashed into a tree, then the probability that the driver was under the influence of alcohol, is

- (1) $\frac{3}{7}$
- (2) $\frac{4}{7}$
- (3) $\frac{5}{7}$
- (4) $\frac{6}{7}$

Q22 - Bayes Theorem

A person goes to office either by car, scooter, bus or train, the probability of which being $\frac{1}{7}$, $\frac{3}{7}$, $\frac{2}{7}$ and $\frac{1}{7}$, respectively. Probability that he reaches office late, if he takes car, scooter, bus or train are $\frac{2}{9}$, $\frac{1}{9}$, $\frac{4}{9}$ and $\frac{1}{9}$, respectively. Given that, he reached office in time, then the probability that he travelled by car, is

- (1) $\frac{1}{5}$
- (2) $\frac{1}{6}$
- (3) $\frac{1}{7}$
- (4) $\frac{1}{8}$

Q23 - Binomial Distribution

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A coin is tossed 7 times. Each time a man call head. The probability that he wins the toss on more than three occasions is

- (1) $1/4$
- (2) $5/8$
- (3) $1/2$
- (4) None of these

Q24 - Binomial Distribution

8 coins are tossed simultaneously. The probability of getting atleast 6 heads, is

- (1) $57/64$
- (2) $229/256$
- (3) $7/64$
- (4) $37/256$

Q25 - Binomial Distribution

If two coins are tossed 5 times, then the probability of getting 5 heads and 5 tails is

- (1) $63/256$
- (2) $1/1024$
- (3) $2/205$
- (4) $9/64$

Q26 - Binomial Distribution

If X is a random variable with distribution given below

X	0	1	2	3
$P(X = x)$	k	$3k$	$3k$	k

Then, the value of k and its variance are

- (1) $1/8, 22/27$

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(2) $1/8, 23/27$

(3) $1/8, 24/27$

(4) $1/8, 3/4$

Q27 - Binomial Distribution

A person can kill a bird with probability $3/4$. He tries 5 times. What is the probability that he may not kill the bird?

(1) $243/1024$

(2) $781/1024$

(3) $1/1024$

(4) $1023/1024$

Q28 - Probability Through Tree Diagram

Miss C has either tea or coffee at morning break. If she has tea one morning, the probability she has tea the next morning is 0.4. If she has coffee one morning, the probability she has tea on the following Wednesday morning is

(1) 0.46

(2) 0.49

(3) 0.51

(4) 0.61

Q29 - Probability Through Tree Diagram

Sixteen players $S_1, S_2, \dots, S_{16}$ play in a tournament. They are divided into eight pairs at random. From each pair, a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength. The probability that 'exactly one of the two players S_1 and S_2 is among the eight winners', is

(1) $4/15$

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(2) $7/15$

(3) $8/15$

(4) $9/15$

Q30 - Probability Through Tree Diagram

The number a is randomly selected from the set $\{0, 1, 2, 3, \dots, 98, 99\}$. The number b is selected from the same set. Probability that the number $3^a + 7^b$ has a digit equal to 8 at the unit place, is

(1) $1/16$

(2) $2/16$

(3) $4/16$

(4) $3/16$

Q4 (3)

Q8 (3)

Q12 (3)

Q16 (2)

Q20 (1)

Q24 (4)

Q28 (2)

Q30 (4)